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Stochastic Dynamic Programming Models for Reservoir Operation Optimization

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Most applications of stochastic dynamic programming have derived stationary policies which use the previous period's inflow as a hydrologic state variable. This paper develops a stochastic dynamic programming model which employs the best forecast of the current period's inflow to define a reservoir release policy and to calculate the expected benefits from future operations. Use of the best inflow forecast as a hydrologic state variable, instead of the preceding period's inflow, resulted in substantial improvements in simulated reservoir operations with derived stationary reservoir operating policies. While these results are for a dam at Aswan in the Nile River Basin, operators of other reservoir systems also have available to them information other than the preceding period's inflow which can be used to develop improved inflow forecasts.

INTRODUCTION

Stochastic dynamic programming (SDP), usually with a Markov chain description of inflows, has often been a useful technique for calculating optimal single-reservoir [*Masse*, 1946; *Little*, 1955; *Buras*, 1963; *Gablinger and Loucks*, 1970; *Loucks and Falkson*, 1970; *Butcher*, 1971; *Dudley and Burt*, 1973; *Torabi and Mobasheri*, 1973; *Su and Deininger*, 1974; *Roefs and Guitron*, 1975; *Bogle and O'Sullivan*, 1979; *Oven-Thompson et al.*, 1982; *Hashimoto et al.*, 1982; *Esmaeil-Beik and Yu*, 1984] and sometimes multiple-reservoir [*Schweig and Cole*, 1968; *Takeuchi and Moreau*, 1974; *Gal*, 1979; *Turgeon*, 1980, 1981] operating policies. Stochastic dynamic programming can derive reservoir operating policies which are subject to reliability constraints [*Mawer and Thorn*, 1974; *Askew*, 1974*a*, *b*, 1975; *Sniedovich and Davis*, 1975; *Rossman*, 1977; *Sniedovich*, 1979; *Klemeš*, 1975]. *Yakowitz* [1982] reviews the evolution of these models. Papers by *Gal* [1979] and *Turgeon* [1980, 1981] are particularly important, for they present stochastic dynamic programming algorithms (parametric-SDP, aggregation-decomposition-SDP, and aggregation-SDP) that can assist in deriving multiple-reservoir operating policies.

Recent research has examined ways of improving stochastic dynamic programming models by improving the model of reservoir inflows [*Verhaeghe*, 1977; *Alarcon and Marks*, 1979; *Maidment and Chow*, 1981; *Bras et al.*, 1983; *Gal*, 1979]. In particular, *Alarcon and Marks* [1979] and *Bras et al.* [1983] employed nonstationary dynamic programming models for the High Aswan Dam. In those studies, the large times of flow within the Nile Basin allowed much more accurate inflow forecasts than those obtainable based on only the preceding period's inflow. *Hoshi and Burges* [1979, 1980] discussed the use of snowpack information to forecast seasonal inflow values; there, too, improved reservoir operating policies should result from use of available hydrologic information beyond that provided by the previous period's inflow. This paper discusses alternative formulations of stochastic dynamic programming models for reservoir operation optimization which can employ efficient flow forecasts as hydrologic state variables. Numerical examples based on the Nile River Basin demonstrate that these optimization models can identify significantly more efficient reservoir operating procedures.

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ALTERNATIVE POLICY-OPTIMIZATION MODELS

Bras et al. [1983] employed a nonstationary "adaptive-control" stochastic dynamic programming algorithm to optimize reservoir operation during the next T periods and beyond where the distributions of inflows during the next T periods were conditioned upon all information available at the beginning of the current period t. After T periods the return function or future benefits corresponding to storage level S_{t+T} and preceding reservoir inflow Q_{t+T-1} was taken to be the steady state solution $f^{ss}(S_{t+T}, Q_{t+T-1})$ to the classic stationary Markov chain inflow SDP reservoir model. Consider initially annual reservoir operation so that each t will represent a year. The steady state undiscounted optimal policy and return function are found by solution of [*Howard*, 1960]

$$f^{ss}(S_t, Q_{t-1}) + g = \max_{R} \; E_{Q_t|Q_{t-1}} \{B(S_t, Q_t, R^*) + f^{ss}(S_{t+1}, Q_t)\} \qquad \forall t \qquad (1)$$

where $E\{\ \}$ denotes expectation over the specified conditional distribution of period t's inflow Q_t given the value of Q_{t-1}, R^* is the actual release equal to the target release R if the physical constraints $S_{\min} \le S_{t+1} \le S_{\max}$ are not violated, $S_{t+1} = S_t + Q_t - R^*$ is the ending storage, Q_{t-1} and Q_t are the preceding and immediate periods' inflows, $B(S_t, Q_t, R^*)$ is the immediate benefit function, and g is the average gain which must be determined [see *Gablinger and Loucks*, 1970]. In practice, (1) is often solved by calculating $f^{ss}(S_t, Q_{t-1})$ at a discrete

set of points and employing a discrete Markov chain approximation of the conditional distribution of Q_t given Q_{t-1}. However, this and other reservoir policy optimization models are described without reference to the discretization eventually employed.

The adaptive-control SDP model formulated at time t based upon information $\mathbf{X}_t$ available at that time can be written as

$$f_{h,t}^{ac}(S_{t+h}, Q_{t+h-1}) = \max_R E\{B(S_{t+h}, Q_{t+h}, R^*) + f_{h+1,t}^{ac}(S_{t+h+1}, Q_{t+h})\} \quad (2)$$

for $h = 0, 1, \cdots, T-1$ and where the expectation at each stage is over Q_{t+h} given both Q_{t+h-1} and $\mathbf{X}_t$. Here $S_{t+h+1} = S_{t+h} + Q_{t+h} - R^*$ describes the dynamics of the system; $h = 0, 1, \cdots, T-1$ describes the decision process at the current ($h = 0$) and future ($h \geq 1$) stages (time periods), and $\mathbf{X}_t$ summarizes all information available at time t which is used to improve subsequent decisions and to calculate the required conditional probability distributions. Again R^* equals R if possible. The required conditional distributions can be calculated as described by *Stedinger* [this issue].

Beyond some time T, knowledge of $\mathbf{X}_t$ has a negligible impact on the conditional distributions of Q_{t+h} given Q_{t+h-1}, so that the reservoir's operation after that time can be described by the steady state model in (1). Thus *Bras et al.* [1983] employed the terminal boundary condition

$$f_{T,t}^{ac}(S_{t+T}, Q_{t+T-1}) = f^{ss}(S_{t+T}, Q_{t+T-1}) \quad (3)$$

enabling solution of (2) by working backwards. *Verhaeghe* [1977] and *Alarcon and Marks* [1979] consider similar nonstationary adaptive SDP algorithms.

Improving the Adaptive-Control Algorithm

One can see that the adaptive-control approach in (2) with (3) is not entirely consistent: when calculating the optimal immediate release at time t using all available information $\mathbf{X}_t$, that procedure describes decisions in each future period $t + h + 1$ as if they will depend only on $\mathbf{X}_t$ and the immediately preceding reservoir inflow Q_{t+h}; however, if the adaptive-control approach is implemented, those future decisions will depend on all flow information (such as $\mathbf{X}_{t+h}$) which will then be available.

Clearly, it would be ideal in one respect to allow decisions in any future period $t + h + 1$ to depend on the entire vector $\mathbf{X}_{t+h}$. Thus one would compute return functions $f(S_t, \mathbf{X}_{t-1})$. The problem with this idea is obvious: the dimension of $\mathbf{X}_{t-1}$ can be large. Thus it is appropriate to summarize the hydrologic information $\mathbf{X}_{t+h}$ that will be available at the beginning of period $t + h + 1$ by a simple surrogate.

An attractive choice for a hydrologic state variable is the minimum variance forecast $\hat{Q}_{t+h+1}$ of the reservoir's inflow during period $t + h + 1$ made using all available information in period $t + h$:

$$\hat{Q}_{t+h+1} = E[Q_{t+h+1} | Q_{t+h}, \mathbf{X}_{t+h}] \quad (4)$$

Our modified nonstationary adaptive-control model can be written as

$$f_{h,t}^*(S_{t+h}, \hat{Q}_{t+h}) = \max_R E_{Q_{t+h}|\hat{Q}_{t+h}} \{B(S_{t+h}, Q_{t+h}, R^*) + E[f_{h+1,t}^*(S_{t+h+1}, \hat{Q}_{t+h+1})]\} \quad (5)$$

with $h = 0, 1, \cdots, T-1$ for each t and where the inner expectation is over $\hat{Q}_{t+h+1}$ given Q_{t+h}, $\hat{Q}_{t+h}$, and also $\mathbf{X}_t$; here $S_{t+h+1} = S_{t+h} + Q_{t+h} - R^*$, and the release R^* equals the target R if possible.

This model includes two conditional expectations at each stage. First, one employs the conditional distribution for Q_{t+h} given $\hat{Q}_{t+h}$; here $\hat{Q}_{t+h}$ is the best forecast of Q_{t+h} which can be made employing $\mathbf{X}_{t+h-1}$, $\mathbf{X}_t$, and possibly other information. Thus this distribution does not depend on the value of $\mathbf{X}_t$ because the influence of that information on the distribution of Q_{t+h} is summarized by $\hat{Q}_{t+h}$ for the linear Gaussian time series models employed here and by *Curry and Bras* [1980]. On the other hand, the inner conditional distribution for the subsequent period's forecast $\hat{Q}_{t+h+1}$ given Q_{t+h} and $\hat{Q}_{t+h}$ in general may depend upon $\mathbf{X}_t$ as well. It is this dependence alone that results in a nonstationary model. If the dependence of $\hat{Q}_{t+h+1}$ upon $\mathbf{X}_t$ is neglected, one obtains the stationary optimization model:

$$f^{**}(S_t, \hat{Q}_t) + g^{**} = \max_R E_{Q_t|\hat{Q}_t} \{B(S_t, Q_t, R^*) + E[f^{**}(S_{t+1}, \hat{Q}_{t+1})]\} \quad \forall t \quad (6)$$

where the inner expectation is over the random variable $\hat{Q}_{t+1}$ given the values of Q_t and $\hat{Q}_t$; here $S_{t+1} = S_t + Q_t - R^*$, R^* equals R if possible, and g^{**} is again the optimal gain. (*Bertsekas* [1976] also considers such models.)

Clearly, (6) is the stochastic model which most reasonably should be employed to specify the final boundary condition for the nonstationary model in (5). However, (6) is itself an attractive alternative to (1), which employs Q_{t-1} as the hydrologic state variable. The stationary policy which results from use of (6) will certainly be as efficient as one developed using (1) because the former would employ in each period t the best available forecast $\hat{Q}$ of Q. Finally, we note that if $E[(\hat{Q}_t - Q_t)^2]$ is small, so that forecasts are very accurate, then (6) is nearly the stationary "perfect-forecast" operating model:

$$f^p(S_t, Q_t) + g^p = \max_{R^*} \{B(S_t, Q_t, R^*) + E_{Q_{t+1}|Q_t} [f^p(S_{t+1}, Q_{t+1})]\} \quad (7)$$

where $S_{\min} \leq S_{t+1} = S_t + Q_t - R^* \leq S_{\max}$. Such models have been used by *Gablinger and Loucks* [1970], *Loucks and Falkson* [1970], *Alarcon and Marks* [1979], *Loucks et al.* [1981], *Hashimoto et al.* [1982], and *Esmaeil-Beik and Yu* [1984]. *Alarcon and Marks* [1979] formulated the adaptive real-time control version of (7).

State and Decision Variable Selection

A key decision in the formulation of the SDP reservoir-operating models has been the choice of either Q_{t-1} or Q_t as a hydrologic state variable. For a weekly SDP reservoir model, *Esmaeil-Beik and Yu* [1984] found that the simulated average weekly losses almost doubled when Q_{t-1} was used to specify the operating policy rather than Q_t. Thus the choice between these state variables has practical implications even with short time steps.

At the beginning of period t, Q_{t-1} should be known, while only a forecast of Q_t would be available. Release policies $R(S_t, Q_{t-1})$, which are a function of initial storage S_t and the previous period's inflow Q_{t-1}, are easy to implement. However, such policies make no attempt to refine the release R in response to the actual flow levels observed during period t unless physical operating restrictions ($S_{\min} \leq S_t \leq S_{\max}$) come into play. Thus policies of the form $R(S_t, Q_{t-1})$ are optimal only among a very restricted class of release rules.

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On the other hand, release policies $R(S_t, Q_t)$, which depend on the actual inflow Q_t which occurs during the period, can often be reformulated in terms of combinations of storage and release targets; then such policies can be implemented without actual knowledge of Q_t at the beginning of period t (see Loucks [1968] and *Loucks et al.* [1981, pp. 331–332] for illustrations). Unfortunately, this is not always possible, as the extreme examples of *Hashimoto et al.* [1982] demonstrate. Still, period t's release R_t is made throughout period t, and Q_t will be known by the end of that period; thus one could strive to implement an $R(S_t, Q_t)$ policy by employing at each moment within period t the best available forecast of Q_t. The question of whether policies which at each moment use the best forecast of Q_t will in practice actually be a better guide to reservoir operators than policies which use only Q_{t-1} remains unanswered. It is certainly possible that policies which assume knowledge of Q_t could in practice perform worse than policies which employ Q_{t-1}. Also, it is very difficult to simulate a continuously updating release policy.

For these reasons the next section of this paper focuses on the improvements which can be achieved using policies which employ information available at the beginning of each period. Still, reservoir operation optimization models that select a single release target R as a function of S_t and any of Q_{t-1}, $\hat{Q}_t$, or even $\mathbf{X}_t$ are optimal only among a restricted class of policies. Operating policies could be defined in terms of final storage targets contingent on acceptable bounds on the release rate. The use of an ending storage target corresponds to use of a rule curve to specify reservoir operation as is commonly done. Use of final storage targets may be the appropriate operating mode when regulating lake levels for recreation or trying to maintain a flood storage zone.

AN EXAMPLE

The Nile River case study described by *Alarcon and Marks* [1979], *Oven-Thompson et al.* [1982], and *Bras et al.* [1983] will be employed here to illustrate the potential advantage of the use of the expected inflow $\hat{Q}_t$ as a state variable instead of the previous period's inflow Q_{t-1}. The drainage area of the Nile River above the High Aswan Dam is some 29,000 km^2. The dam has a storage capacity of 169×10^9 m^3 and a height of 110 m. The natural mean annual flow at Aswan is some 84×10^9 m^3. The simulation model included withdrawals by Sudan amounting to 18.5×10^9 m^3 per year.

Table 1 of *Bras et al.* [1983] reports the mean, standard deviation, and historical maximum and minimum of the monthly flows at Wadi Halfa gage (1912–1965). However, the month-to-month correlations reported in Bras et al.'s Table 1 do not match the values we obtained for those statistics, even though all the other statistics matched; see our Table 1.

TABLE 1. Lag One Correlation for Streamflows at Wadi Halfa for the 1912–1965 Period

Month	Sample Lag One Correlation (r_1)	Lag One Correlation Reported by *Bras et al.* [1983]
Jan.	0.257	0.486
Feb.	0.908	0.276
March	0.750	0.555
April	0.865	0.640
May	0.871	0.471
June	0.795	0.270
July	0.496	0.607
Aug.	0.638	0.805
Sept.	0.712	0.896
Oct.	0.785	0.887
Nov.	0.807	0.929
Dec.	0.895	0.874

TABLE 2. Precision of Various Forecast Equations: Square of Lag One Correlation of Monthly Flows $(r_1)^2$, Unadjusted R^2 for our Forecast Equations, Unadjusted R^2 for Curry and Bras' Forecast Equations, and Square of Correlation of our Flow Predictor $\hat{Q}_t$ and Previous Flow Q_{t-1}, All Based on Extended 1906–1967 Record

Month	$(r_1)^2$	R^2 for Our Predictor	R^2 for *Curry and Bras* [1980] Predictor	Squared Correlation Between $\hat{Q}_t$ and Q_{t-1}
Jan.	0.137	0.970	0.991	0.143
Feb.	0.857	0.961	0.987	0.889
March	0.567	0.717	0.991	0.776
April	0.653	0.741	0.984	0.906
May	0.755	0.927	0.968	0.821
June	0.643	0.871	0.897	0.738
July	0.242	0.472	0.663	0.546
Aug.	0.373	0.695	0.810	0.561
Sept.	0.493	0.652	0.662	0.785
Oct.	0.651	0.653	0.718	1.000
Nov.	0.672	0.787	0.850	0.832
Dec.	0.746	0.886	0.973	0.856

The reservoir operation optimization models in (2), (5), and (6) employ either a multivariate streamflow model or one-step-ahead flow predictions. *Curry and Bras* [1980] developed a multivariate monthly streamflow model for the Nile. However, we felt uncomfortable with their predictive equations because the variables included in the models did not always seem consistent with the time of flow between the various gages. An alternative set of predictors were developed which are more physically plausible to us though no more statistically valid [see *Sule*, 1984].

Table 2 reports the square of the lag one correlation coefficient of the flows for the extended 1906–1967 period and the unadjusted R^2 values for the predictors developed by *Sule* [1984] and *Curry and Bras* [1980]. The unadjusted R^2 values for our predictors are substantially larger than the lag one correlations squared ($r_1{}^2$). On the calibration data set, the unadjusted R^2 values obtained by *Curry and Bras*' [1980] predictors are even larger than those obtained by the models we constructed.

For the steady state "predictive" dynamic programming model in (6), the transition probabilities for the discretized monthly flows Q_{jt} following $\hat{Q}_{it}$ (t denotes a month) and of $\hat{Q}_{k,t+1}$ following Q_{jt} (neglecting the dependence upon $\hat{Q}_t$, which was minor given the actual value of Q_t) were derived assuming that the two variables of concern had a bivariate normal distribution with their historical correlations given in Table 2. The discrete values of all random variables were chosen to be the 97.5, 83.125, 59.375, 35.625, and 11.875 percentiles of the unconditional distributions; these represent the medians (as opposed to midpoint) of the interval defined by the quantiles corresponding to unconditional probability levels of 1.00, 0.95, 0.7125, 0.4750, 0.2375, and 0.00, as suggested by *Alarcon and Marks* [1979]. Transition probabilities for the monthly flows in a discrete version of (1) were calculated using the same procedure and percentage points.

A key parameter in this and Bras et al.'s study is the objective function used to derive the various operating policies and in the evaluation by simulation of their operation with the 1912–1965 historical flows. Let R_{ty} be the reservoir release in month t of year y; let PG_{ty} be the power generated by that release. Then the monthly losses due to failure to meet flood

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control, irrigation, and power generation targets were

$$L_{ty} = 10^7[\text{Max}\ (0, I_t - R_{ty})]^2 + 10^4[\text{Max}\ (0, R_{ty} - R_{\max})]^2 + 10^{-3}[\text{Max}\ (0, PT - PG_{ty})]^2 \quad (8)$$

Here the weights (10^7, 10^4, and 10^{-3}) were selected by *Bras et al.* [1983] to reflect the priority on the minimum monthly irrigation releases I_t [see *Bras et al.*, 1983, Figure 4], the maximum desired monthly release rate $R_{\max} = 7.6 \times 10^9$ m^3/month, and the monthly power generation target PT. Bras et al. took $PT = 1280$ GWh per month. The effective installed monthly capacity is more nearly 1260 GWh per month. Given the average annual flow in the river, the average power generation level can hardly exceed 700 GWh/month. Thus, even if no irrigation and maximum flow target violations occur, average annual loss due to $PG_{ty} < PT$ will run 4040 with $PT = 1280$ GWh/month and will run 1080 and 30 were PT decreased to 1000 or 750 GWh/month, respectively.

Alternative Models

Table 3 summarizes the simulated monthly operation of the High Aswan Dam with five different policies derived with various SDP algorithms.

Model SSDP0. This is the tradiational stationary SDP model without discounting (see (1)) in which the state variables are the initial storage level and previous period's inflow volume. *Buchanan and Bras* [1981] discuss the use of this model and simulated the policy they derived to obtain the results reported here. Because the simulation exercises reported here and by *Bras et al.* [1983] do not discount the reservoir operating costs (i.e., they consider only average annual operating costs and energy generation levels), reservoir operation optimization models which strive to optimize the average (undiscounted) operating costs are considered.

Model SSDP1. This was essentially Buchanan and Bras' stationary SDP algorithm. However, the month-to-month flow transition probabilities were calculated using the historical lag one correlation coefficients we obtained with their data. Model SSDP0 was based on, we assume, the lag one correlations reported by *Bras et al.* [1983] and the last column in Table 1.

Model ADP. This is the adaptive-control SDP model, equations (2)–(3), for undiscounted cost optimization developed by *Buchanan and Bras* [1981]. The values listed in Table 3 are those they reported. *Bras et al.* [1983] discuss the discounted cost, adaptive-control SDP version of this model.

TABLE 3. Summary of Simulation Results for the Five Stochastic Operating Policies

	Models Used to Derive Operation Policy				
	SSDP0	SSDP1	PDP1	ADP	PDP2
Average annual cost	5735	5381	5330	5083	4878
Number of months with excess releases	10	10	8	5	1
Average excess release (10^9 m^3/month)	8.09	7.91	7.89	7.85	7.65
Average monthly energy (GWh/month)	679	680	680	681	681

For the SSDP1, PDP1, and PDP2 models the standard errors of estimate for annual cost are 206, 198, and 173; for monthly energy generated they are 9, 9, and 12. Results for the SSDP0 and ADP models were taken from *Buchanan and Bras* [1981], who do not report standard errors. In none of these simulations using the 1912–1965 historical flows did irrigation deficits occur.

Model PDP1. This is the "predictive" stationary SDP model in (6) with the dependence of $\hat{Q}_{t+1}$ upon $\hat{Q}_t$ omitted in the inner expectation. This model employed our one-step-ahead forecasts which, over the historical period used to derive the forecast models' parameters and to simulate the reservoir policies, had lower R^2 values than the predictors derived by *Curry and Bras* [1980].

Model PDP2. This is our predictive SDP algorithm (model PDP1) except that it employs the one-step-ahead forecast equations derived by *Curry and Bras* [1980]; see Table 2 and associated discussion.

Numerical Results

As seen in Table 3, while SSDP0 and SSDP1 are the same model structurally, use of the historical lag one correlations in SSDP1 allowed it to achieve a 6.2% reduction in average operating costs. The PDP1 model's policy performed only marginally better than the policy obtained with the SSDP1 algorithm. However, both achieve near-perfect operation. Both avoid irrigation deficits and substantial floods while achieving roughly the maximum possible annual hydropower generation level. The adaptive-control policy developed by the ADP model did slightly better than policies derived with the SSDP1 and PDP1 models. However, the predictive DP algorithm using the forecast equations developed by *Curry and Bras* [1980], denoted PDP2, generated a policy which performed even better than the more sophisticated ADP model's optimal policy, which also employed Curry and Bras' predictive equations.

Before drawing conclusions from these results, note that the standard error of estimate on the reported average annual losses in Table 3 is about 200 units (see table footnote). Moreover, in *Buchanan and Bras'* [1981] report, their undiscounted loss SSDP model achieved average annual losses of 5735, while the discounted loss minimization version of that same model generated a policy which achieved average annual losses of only 5315. Those results and our own analysis indicate that the differences among losses reported in Table 3 are too small and sensitive to a few key flood events in the historical record to justify a strict ranking of the models. Having simulated the policies with a subset of the flows used to derive the forecasts compounds the problem. Moreover, the observed differences in simulated annual costs are mostly due to flood control operations; these models with their coarse discretization of the flow distributions are not really appropriate for fine tuning the Aswan Dam's flood control policy. Certainly, discrete points at the unconditional continuous distributions' 97.5 and 83.1 percentiles with unconditional probabilities of 5% and 23% provide a crude description of flood potential and flood risk. Before such policies are put into actual use, use of a finer description of the distribution of flood flows in the critical flood months seems imperative. Use of parametric stochastic dynamic programming [*Gal*, 1979] or other techniques which can describe the continuous distribution of the flows [e.g., *Bogle and O'Sullivan*, 1979] provide other alternatives.

Other Examples

The example in Table 3 resulted in little distinction between the operation of the various policies because they achieved near-perfect system operation. To illustrate the potential advantage of use of $\hat{Q}_t$ instead of Q_{t-1} as a hydrologic state variable, the original Aswan Dam operating problem was modified. We lowered the energy target which was the ef-

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fective contract energy level because power generated above the target was assigned zero value. Table 4 describes the simulated operation of policies derived with energy targets of $PT = 1000$ GWh/month and 750 GWh/month.

To illustrate further the potential advantage of the predictive SDP model, the monthly power target was left at $PT = 750$ GWh/month, the flood control objective was dropped, and the dam's capacity was artificially decreased from 169×10^9 m^3 to 127.5×10^9 and 85×10^9 m^3. The performance of policies derived with the SSDP and PDP algorithms for these cases is summarized in Table 5. The annual average loss achieved by the PDP policies employing $\hat{Q}_t$ are 20% and 18% less than those achieved by the SSDP model's policies. These differences are not affected by capricious flood flows and reflect improved irrigation and hydropower generation operations when reservoir and streamflow levels were unusually low; when flow and reservoir levels were high, both policies strived to make only the largest useful release.

Other Applications and Extensions

The vast extent of the Nile River system allows development of extended forecasts of future inflows which are more precise than those obtained using only the inflow to the dam in the previous period. In general, the preceding period's inflow Q_{t-1} need not be the best hydrologic state variable. The preceding period's inflow explicitly describes the past without any explicit attempt to specify what may happen in the future, given the information available to reservoir operators at the beginning of each reservoir operating period. Such information may arise in several forms.

For example, *Hirsch* [1979] and *Hipel et al.* [1977, 1979] suggest that the persistence of monthly streamflows is better described by monthly autoregressive moving average (ARMA) (1,1) models than by the Thomas-Fiering or monthly AR (1) model implicit in (1). Use of the best forecast of the current period's flow $\hat{Q}_t$ as a state variable in (6) would allow identification of policies based on a monthly ARMA (1,1) flow model; in this instance, the inner expectation over $\hat{Q}_{t+1}$ is not even required because $\hat{Q}_{t+1}$ can be written as a linear function of Q_t and $\hat{Q}_t$. An approximate description of other ARMA (p, q) models could also be employed.

However, the largest gains from use of predicted inflows $\hat{Q}_t$ or some other statistic as a state variable will probably be realized in cases where information is available to supplement the streamflow record at a dam. *Hoshi and Burges* [1979, 1980] discuss the use of snowpack information to forecast total seasonal runoff. For such cases the water content of the residual snowpack might be the appropriate hydrologic state variable for modeling the decision and hydrologic processes

TABLE 4. Average Annual System Operating Loss for Different Monthly Energy Targets

Monthly Energy Target, GWh/Month	Models Used to Derive Operations Policy		Difference, %
	SSDP	PDP	
1280	5381	5330	1.0
1000	2343	2213	5.9
750	1040	786	32.0

For the SSDP models, the standard errors of estimate are 206, 288, and 381; for the PDP models they are 198, 304, and 226 for targets of 1280, 1000, and 750 GWh/month, respectively.

TABLE 5. Simulation Results for Policies Derived With Different Reservoir Capacities and a Monthly Energy Target of 750 GWh/Month and No Flood Control Objective

	Reservoir Capacity, 10^9 m^3			
	85*		127.5†	
	SSDP	PDP	SSDP	PDP
Average annual loss	3.45×10^6	2.76×10^6	749	611
Number of irrigation deficits	69	68	3	1
Average annual irrigation deficits, 10^6 m^3	344	295	1.0	0.2
Average monthly energy, GWh	530	531	581	581

*For the SSDP and the PDP models the standard errors of estimate for the annual loss are 2.74×10^5 and 2.32×10^5; for the monthly energy generated they are 8 in both cases.

†For the SSDP and the PDP models the standard errors of estimate for the annual loss are 175 and 41; for the monthly energy generated they are 5 in both cases.

and for specifying monthly reservoir operating rules during the spring snowmelt period.

In the Nile example, streamflow levels observed throughout the basin provided supplemental information. The historical records at those sites were used to develop the monthly streamflow models subsequently used for predicting inflows to the dam. Other types of streamflow forecast models will sometimes be appropriate. The time of flow from various upstream gages to Lake Nasser behind the Aswan Dam was seldom an integral number of months. As a result, to obtain precise estimates of the inflows to the dam during the next 30 days, reservoir operators could employ observed daily flow rates throughout the basin with forecasts of future daily flow levels in a deterministic flow routing algorithm. Such an approach based upon the physical characteristics of a basin and observed flow rates is attractive. More accurate predictions may be obtained, and long streamflow records are not required to estimate the parameters of the routing models.

Conclusion

The early work by *Masse* [1946], *Little* [1955], *Buras* [1963], and *Butcher* [1971] and most subsequent applications of stochastic dynamic programming have derived stationary policies which use the previous period's inflow as a hydrologic state variable. Recently, *Alarcon and Marks* [1979] and *Bras et al.* [1983] used nonstationary SDP models. Their algorithms would be rerun each month to derive an optimal release for that month based on the conditional distribution of future inflows. While these nonstationary procedures are potentially more efficient than stationary policies, they do have their drawbacks. In planning studies, the computational cost of rerunning the SDP model each month to find the optimal release for that month will probably make the use of such procedures prohibitively expensive in modest Monte Carlo studies of even single-reservoir systems. Likewise, the cost of experimenting with the policies implicitly defined by such sophisticated optimization procedures is likely to discourage their actual adoption for reservoir system operation.

The point stressed here is that both stationary and nonstationary SDP models can identify better reservoir operating

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policies if better hydrologic state variables are employed. By using *Curry and Bras'* [1980] forecast equations for the current period's inflow, a stationary reservoir operating policy was obtained that outperformed the corresponding nonstationary reservoir operating algorithm which employed the preceding period's inflow as the hydrologic state variable. Tables 4 and 5 show that substantial improvements can be obtained in reservoir operations with stationary reservoir operating policies which use a one-month-ahead flow forecast as a hydrologic state variable. While these results are for a dam at Aswan in the Nile River Basin, operators of other reservoir systems also have available to them information which can be used to develop significantly improved inflow forecasts.

The results here illustrate the potential value of snowpack, lake level, soil moisture, groundwater, and streamflow information likely to be available to improve reservoir operations. Stochastic dynamic programming algorithms only identify the optimal policy within the class of policies examined; if policies in the space examined fail to make use of such hydrologic information, or employ only release targets when ending storage targets with constraints on releases would be more appropriate, then the best reservoir operating policies cannot be identified.

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